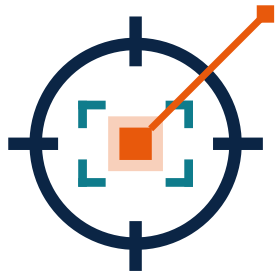
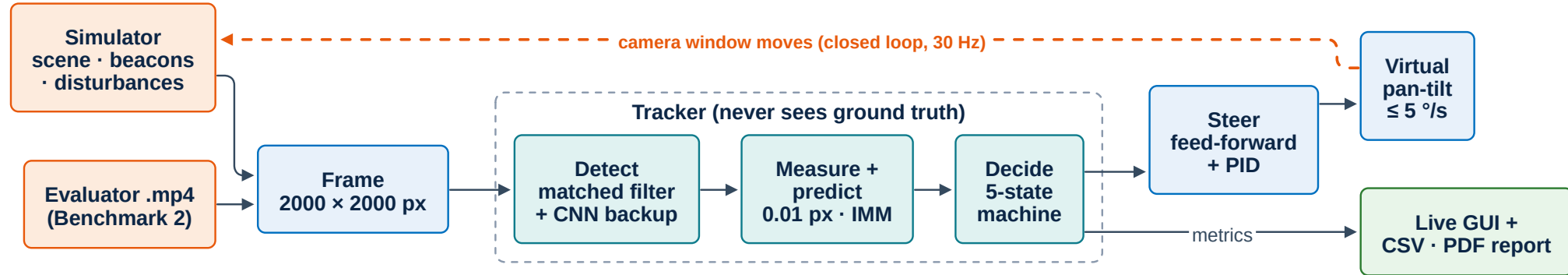


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AI virtual camera tracker for coarse alignment of mobile FSOC terminals

- **Proposed Solution** A standalone desktop app (plus the same program as a web app) that **renders a 2000×2000 px sky with a moving beacon and real-world disturbances**, then detects, measures, predicts and steers a rate-limited 640×480 virtual camera to keep that beacon centred. Every run writes its own performance report. Evaluator .mp4 files plug into the same pipeline.



■ How it addresses the problem

OBSERVE

Tracker watches the whole 2000×2000 px (12.5°) scene each frame, as the PS asks (“observe the environment”); hard mode limits it to the window.

ACQUIRE

Matched filter + noise-adaptive threshold; a spot must hold 3 of 4 frames before lock. Locks in 0.7 to 1.4 s (spec ≤ 2 s).

ESTIMATE

Centre to ≈ 0.01 px by 2D Gaussian fit; an IMM filter predicts line, circle, figure-8 and random paths unaided.

ADJUST

Feed-forward + PID leads the beacon by the gimbal's 1-frame delay, never above 5 °/s: 6.4 to 8.2 px error (spec ≤ 10 px).

■ Innovation and uniqueness

Physics-ordered disturbances

Atmosphere, then platform sway and jitter, then sensor noise: the order real light meets them.

Self-trained AI safety net

84 k-parameter CNN trained on our own simulator frames with exact labels; no dataset needed.

Provably blind to the truth

Ground truth deleted before tracking gives identical results to 10^{-6} px.

Track-before-detect

Links 3 to 6 σ spots across frames. Faint-beacon lock went from never to 91–97.5 %.

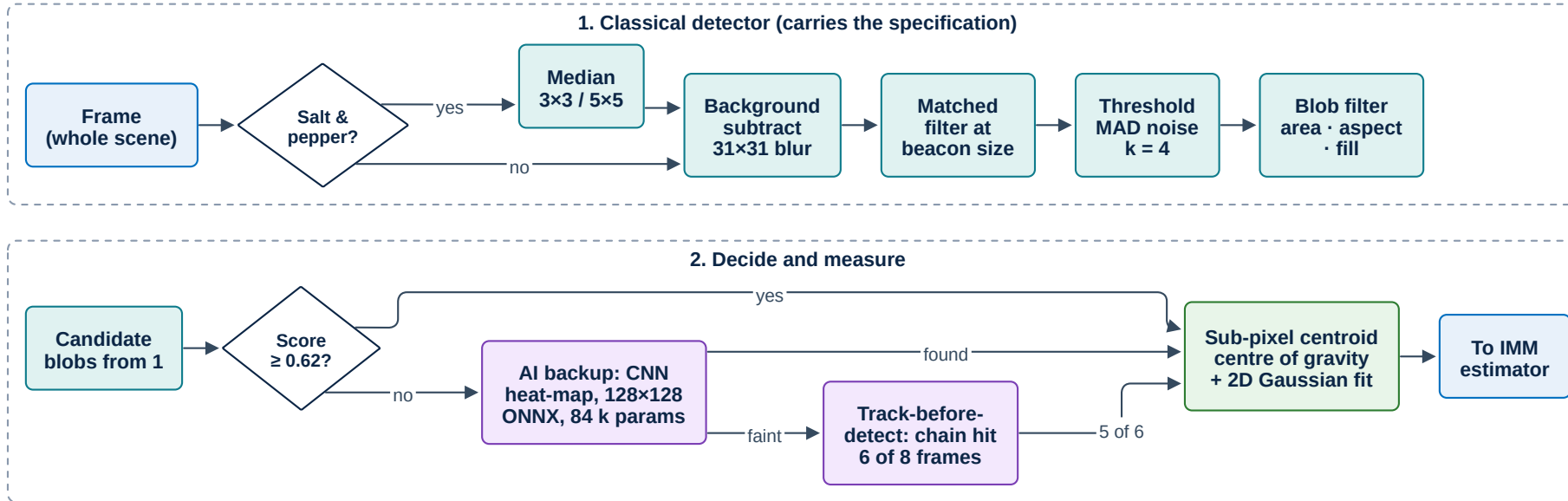
One pipeline, two sources

An evaluator .mp4 replaces the simulator; the tracker code runs unchanged (Benchmark 2).

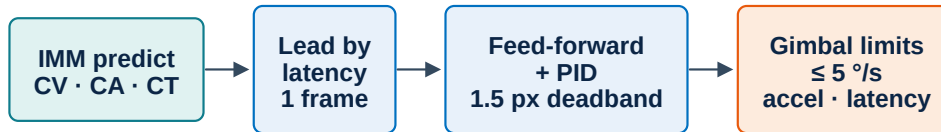
Holds identity among decoys

Named targets; appearance, start cue or click picks the one to follow, re-checked every 0.5 s.

Every frame: detect and measure



Then estimate and steer


22.5"

one pixel (4° / 640 px)

26.7 px

max slew per frame at 5 °/s, 30 Hz

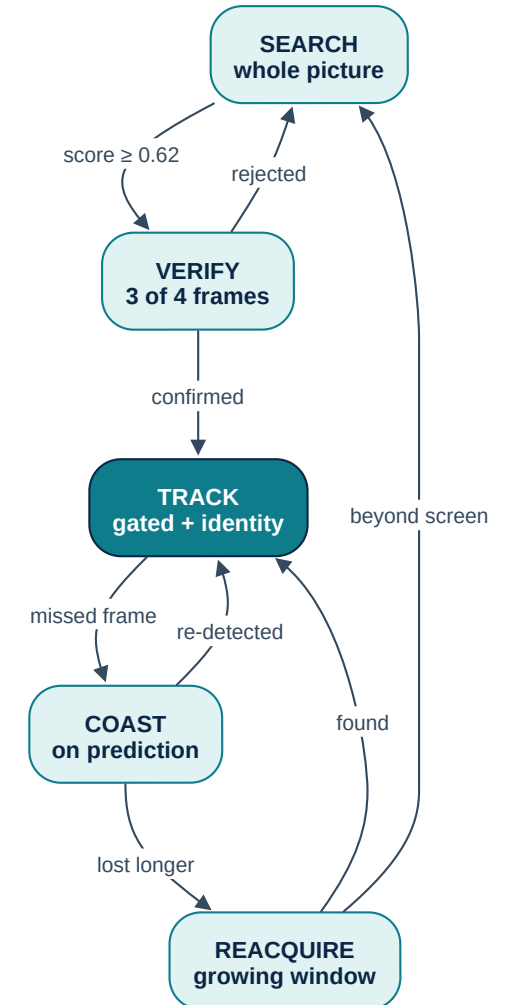
0.95 s

centre to corner, target known

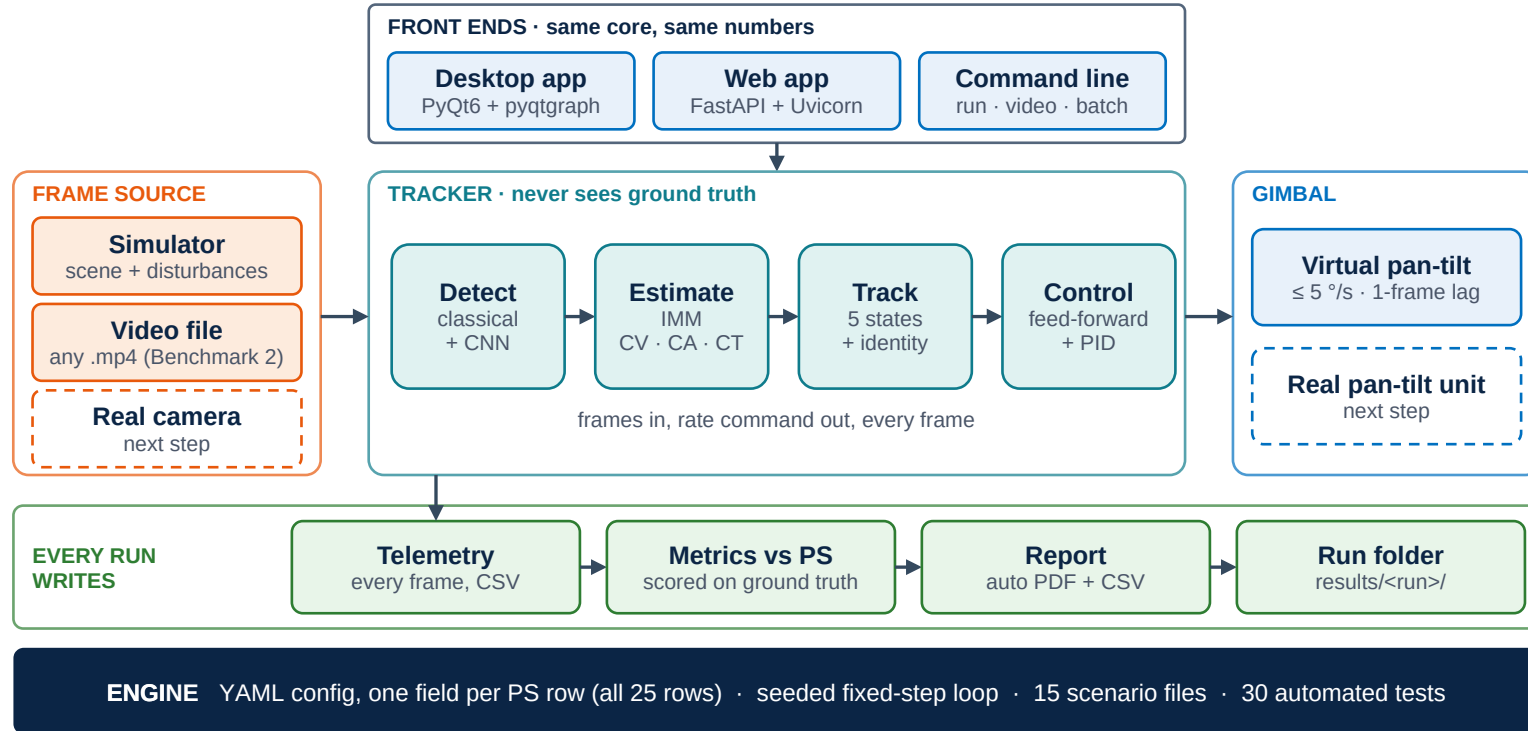
50 ms

budget at 20 FPS; we need 4–12 ms

Tracker states



Software architecture



Technologies used

LANGUAGE + CORE



COMPUTER VISION



AI



DESKTOP GUI



WEB APP



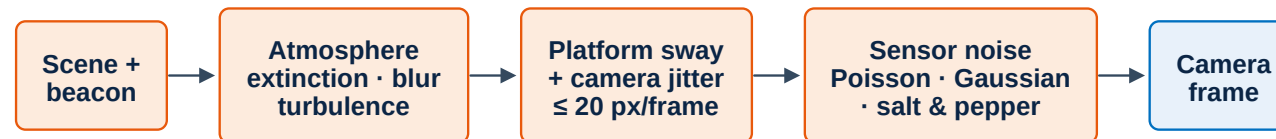
REPORTS

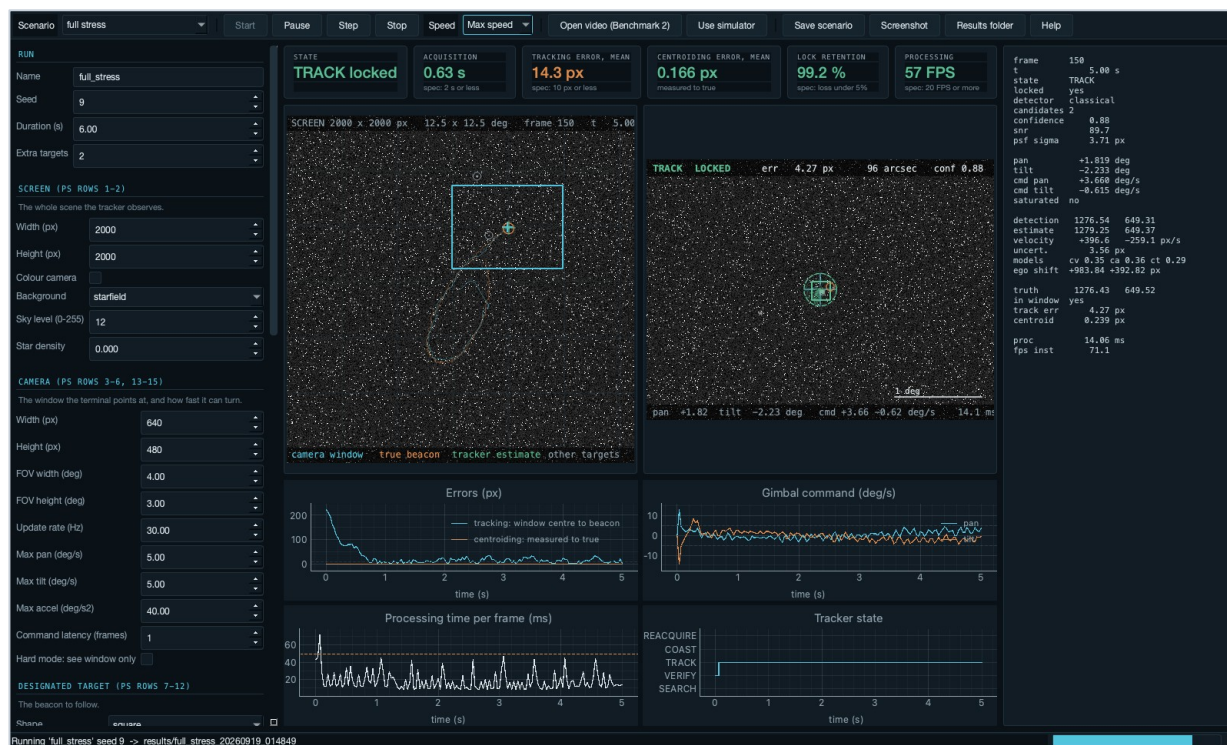


BUILD + QA



Disturbances applied in physical order





Desktop app mid-run, multi-target stress scenario (haze, 3 noise types, sway, jitter, 2 decoys): TRACK locked, 0.63 s acquisition, 99.2 % lock, 57 FPS with live drawing; 14.3 px error is the injected shake

Results on clear runs against the PS targets

0.7–1.4 s

acquisition
spec ≤ 2 s

6.4–8.2 px

tracking error
spec ≤ 10 px

100 %

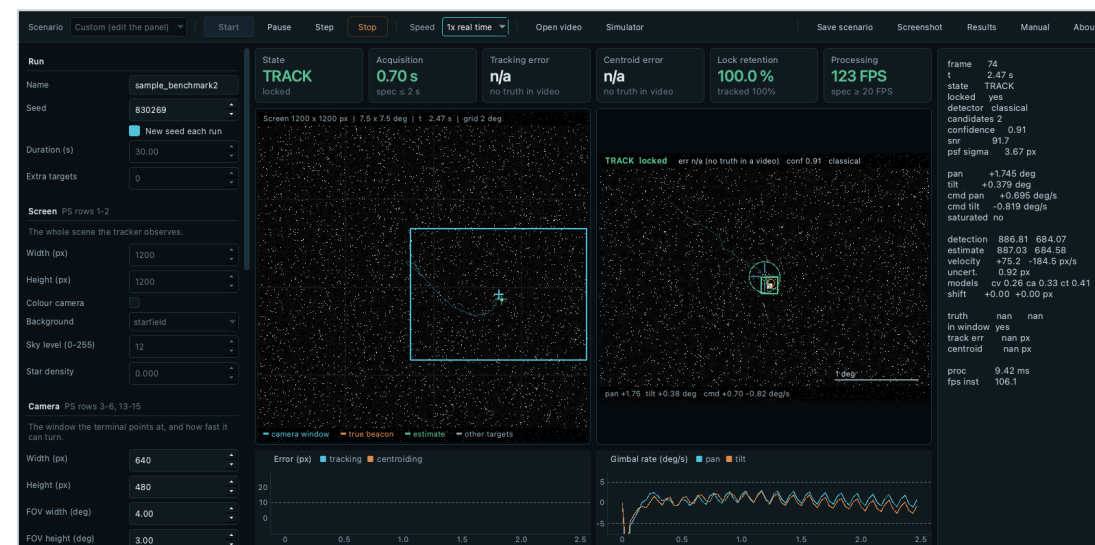
lock retention
spec: loss < 5 %

0.006 px

centroid error, clear
0.1–0.4 px heavy noise

80–250

FPS on a laptop
spec ≥ 20 FPS



Benchmark 2 input: sample_benchmark2.mp4 (1200 × 1200 px, 30 fps) opened directly. The simulator is bypassed; size, frame rate and rotation come from the file. Our measured beacon centre is logged every frame (det_x, det_y) to compare with the predefined values.

Source code: github.com/tanmayhutt/SIH26169 (private; access on request)

Web app: sih26169.blankpoint.club (login)

Demo video (≈ 4 min): sih26169.blankpoint.club/downloads (login)

■ Feasibility: already built and measured

Scenario (15 s runs, 3 seeds)	Acquire	Error	Lock	Verdict
Clear: line, circle, figure 8	0.7–1.4 s	6.4–8.2 px	100 %	Pass
Clear random walk	≈ 1.1 s	≈ 10 px	98 %	At limit
Heavy noise (S&P 10 %, σ 20, Poisson)	0.8–1.1 s	7.3–8.2 px	100 %	Pass
Fog and low light	0.7–1.7 s	6.6–8.0 px	100 %	Pass
Faint beacon, 10 seeds	1.3–5.5 s	6.7–12.3 px	91–97.5 %	Most seeds
Decoys + haze + noise + sway	0.7–1.3 s	13.6–14.4 px	98–99 %	Identity held
Platform at PS max (20 + 20 px/frame)	≈ 0.8 s	≈ 25 px	76–88 %	Physical limit
Hard mode (camera window only)	3–12 s	7–12 px	100 %	By design

Viability: runs offline on an ordinary laptop CPU, no camera, pan-tilt unit or optics to buy. Same numbers on Windows, Linux and both Macs; every build ships only after its own packaged smoke test.

■ Challenges, risks and our strategy

Challenge or risk	Our strategy
Random shake at the PS maximum (20 px/frame) cannot be predicted	Report error with shake removed beside the raw number; a 10 °/s motor was tested and does not help
Unseen evaluator videos (60 % of marks)	Never tune to our own clips; read true size, fps and rotation from each file
Beacon too faint for any single frame	Track-before-detect: chain weak spots over 6 of 8 frames, then confirm 5 of 6
Decoys crossing the designated beacon	Designate by appearance, a start cue or a click; identical look-alikes: 100 % lock with the start cue
An AI model failing silently or slowly	Classical path carries the spec; CNN only fills gaps, loaded and warmed up at start

■ How every change is verified before it is accepted



WHY IT MATTERS NOW

Optical links are already here, and every one starts with coarse pointing.

200 Gbps

NASA TBIRD, space to ground, 2023 [11]

20 Gbps · 20 km

Taara FSO links in Airtel's network, India [15]

1.8 Gbit/s

ESA EDRS relay to satellites 35,000+ km away [13]

Who it helps**ISRO SAC and DoS PAT teams**

Try coarse-pointing algorithms before a camera or gimbal is bought.

**Universities and students**

A free, repeatable PAT lab: the PS's own "development and learning" goal.

**Indian FSOC, UAV and NewSpace start-ups**

Benchmark trackers on the same scenarios and per-frame logs.

**Defence and disaster communications**

Rapid optical links between vehicles and drones need a fast coarse lock.

What it brings**Economic**

One laptop replaces a camera, pan-tilt and optics bench.

**Technical**

Seeded, repeatable runs; per-frame CSV makes trackers comparable.

**Social**

Offline on Windows, Linux and macOS; nothing to install.

**Environmental**

Fewer hardware prototypes; light needs no spectrum licence.

**Strategic**

Indigenous PAT software for India's optical and quantum links [14].

**Hardware-ready**

Camera and gimbal are interfaces, so real ones plug straight in.

Path from virtual bench to a real link

■ FSOC and pointing, acquisition and tracking

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■ Tracking, estimation and vision methods

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■ Missions and deployments

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- [12] NASA, "NASA's laser comms demo makes deep space record, completes first phase" (DSOC), 2024.
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■ Software and project documents

- [16] ONNX Runtime; G. Bradski, "The OpenCV Library," Dr. Dobb's J. Softw. Tools, 2000.
onnxruntime.ai
- [17] SIH 2026 Problem Statement 26169, ISRO Space Applications Centre (parameter table and evaluation method, rows 1 to 25).
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■ Coverage, definitions and known limits

Coverage: all 25 PS table rows, 8 of 8 "shall" functions, all 5 deliverables and the optional video.

Performance log, every run: duration, FPS, acquisition and re-acquisition time, mean, max and RMSE tracking and centroiding error, lock retention, processing time; a CSV row per frame.

Definitions: lock = TRACK state within 30 px of the camera centre; tracking error = true beacon to camera centre; centroiding error = our measured centre to the true centre (the PS leaves these open).

Known limits: one faint-beacon seed in 10 acquires in 5.5 s; hard mode searches for 3 to 12 s; a faint beacon that never moves is not covered; the builds are unsigned, so the OS warns on first launch.